Т3

$$p(x)=\begin{cases}\dfrac{\exp\left[-\frac{x}{\theta}\right]}{\theta}, & x\geq 0\\ 0, & x<0\end{cases} \qquad \theta>0.$$

• Рассмотрим $\tilde{\theta}_1$:

$\tilde{\theta}_1=\overline{x}$

$$M(\tilde{\theta}_1)=M\left[\frac{1}{n}\sum_{i=1}^{n}x_i\right]=\left\{\begin{matrix}\text{независимы,}\\ \text{одинаково}\\ \text{распред.}\end{matrix}\right\}=$$

$$=\frac{1}{n}\sum_{i=1}^{n}M[x_i]=M\xi=\theta.$$

$\Rightarrow \tilde{\theta}_1$ – несмещенная.

$$D[\tilde{\theta}_1]=D\left[\frac{1}{n}\sum_{i=1}^{n}x_i\right]=\frac{1}{n^2}D\xi=\frac{\theta^2}{n}$$

$n=3$: $D[\tilde{\theta}_1]=\dfrac{\theta^2}{3}$

• Рассмотрим $\tilde{\theta}_2$:

$\tilde{\theta}_2=x_{(2)}$

$$\varphi(y)=n(n-1)\left(1-e^{-y/\theta}\right)\left(e^{-y/\theta}\right)^{n-2}\cdot\frac{1}{\theta}e^{-y/\theta}=$$

$$=\frac{n(n-1)}{\theta}\left(e^{-\frac{n-1}{\theta}y}-e^{-\frac{n}{\theta}y}\right).$$

$$M[\tilde{\theta}_2] = \int_0^{+\infty} y\varphi(y)dy = \frac{n(n-1)}{\theta}\int_0^{+\infty} y\left(e^{-\frac{n-1}{\theta}y} - e^{-\frac{n}{\theta}y}\right)$$

$$dy = \left\{\int_0^{+\infty} y e^{-ay}dy = \frac{1}{a^2}\right\} = \frac{n(n-1)}{\theta}\left(\frac{\theta^2}{(n-1)^2} - \frac{\theta^2}{n^2}\right) =$$

$$= \theta\frac{n(n-1)(n^2-(n-1)^2)}{n^2(n-1)^2} = \theta\cdot\frac{2n-1}{n(n-1)}$$

$M[\tilde{\theta}^2] \neq \theta \Rightarrow \tilde{\theta}_2$ – смещенная.

$$\tilde{\theta}_2' = \frac{n(n-1)}{2n-1}\tilde{\theta}_2 = \frac{n(n-1)}{2n-1}x_{(2)}$$

$\tilde{\theta}_2'$ – несмещенная

$n=3$: $\tilde{\theta}_2' = \frac{6}{5}x_{(2)}$

• Сравним эффективности оценок $\tilde{\theta}_1$ и $\tilde{\theta}_2'$:

$$D[\tilde{\theta}_1] = \frac{\theta^2}{3}$$

$$M[\tilde{\theta}_2^2] = \int_0^{+\infty} y^2\varphi(y)dy = \int_0^{+\infty} y^2\cdot\frac{n(n-1)}{\theta}\left(e^{-\frac{n-1}{\theta}y} - e^{-\frac{n}{\theta}y}\right)$$

$$dy = \frac{n(n-1)}{\theta}\int_0^{+\infty} y^2\left(e^{-\frac{n-1}{\theta}y} - e^{-\frac{n}{\theta}y}\right)dy =$$

$$= \left\{t = \frac{y}{\theta}n\right\} = \theta^2\left(\frac{n}{n^2-n}\int_0^{+\infty} t^2e^{-t}dt - \frac{n-1}{n^2}\int_0^{+\infty} t^2e^{-t}dt\right) =$$

$$= 2\theta^2\left(\frac{n^3-(n-1)^3}{n^2(n-1)^2}\right) = 2\theta^2\cdot\frac{(3n^2-3n+1)}{n^2(n-1)^2}$$

$$D[\tilde{\theta}_2] = M[\tilde{\theta}_2^2] - (M[\tilde{\theta}_2])^2 = \theta^2 \cdot \frac{(2n^2-2n+1)}{n^2(n-1)^2}$$

$$D[\tilde{\theta}_2'] = \left(\frac{n(n-1)}{2n-1}\right)^2 D[\tilde{\theta}_2] = \theta^2 \cdot \frac{(2n^2-2n+1)}{(2n-1)^2}$$

$n=3$: $D[\tilde{\theta}_2'] = \frac{13}{25}\theta^2$

$\frac{1}{3}\theta^2 < \frac{13}{25}\theta^2 \Rightarrow \tilde{\theta}_1$ более эффективна.

- $p(x)$ – непр. дифф. по θ на $(0;+\infty)$.
- $\frac{\partial}{\partial\theta}\left(\int_0^{+\infty} \frac{1}{\theta}\exp\left[-\frac{x}{\theta}\right]dx\right) = \frac{\partial}{\partial\theta}(1) = 0$
- $\int_0^{+\infty}\left(\frac{x}{\theta^3}\exp\left(-\frac{x}{\theta}\right) - \frac{1}{\theta^2}\exp\left(-\frac{x}{\theta}\right)\right)dx = \frac{1}{\theta} - \frac{1}{\theta} = 0$

$$\Rightarrow \frac{\partial}{\partial\theta}\int_0^{+\infty} p(x,\theta)\,dx = \int_0^{+\infty}\frac{\partial}{\partial\theta}p(x;\theta)\,dx$$

- $\ln\left(\frac{1}{\theta}\cdot\exp\left[-\frac{x}{\theta}\right]\right) = -\frac{x}{\theta} - \ln\theta$

$$\left(\frac{\partial \ln p(x,\theta)}{\partial\theta}\right)^2 = \left(\frac{x}{\theta^2} - \frac{1}{\theta}\right)^2$$

$$I(\theta) = M\left[\left(\frac{\partial \ln(p(x,\theta))}{\partial\theta}\right)^2\right] =$$

$$= \int_0^{+\infty}\left(\frac{x^2}{\theta^2} - \frac{2x}{\theta^3} + \frac{1}{\theta^2}\right)\cdot\frac{1}{\theta}\cdot\exp\left(-\frac{x}{\theta}\right)dx =$$

$$= \frac{1}{\theta^2}\left(\int_0^{+\infty} t^2 e^{-t}\,dt - 2\int_0^{+\infty} t e^{-t}\,dt + \int_0^{+\infty} e^{-t}\,dt\right) =$$

$$= \frac{1}{\theta^2}\left(2-2+1\right) = \frac{1}{\theta^2}$$

$I(\theta)$ - непр.

$I(\theta) > 0$ на $(0;+\infty)$

$\Rightarrow$ вер. модель регулярна

$\tilde{\theta}_1$:

- Несмещенная
- ВМ-регулярна
- $D\tilde{\theta}_1 = \frac{\theta^2}{n}$ - огр. на $\forall$ компакте из $(0; +\infty)$

$\Rightarrow$ $\tilde{\theta}_1$ - регулярная оценка.

$\tilde{\theta}_2$:

- Несмещенная
- ВМ - регулярна

$$\cdot D\tilde{\theta}_2' = \theta^2 \cdot \left(\frac{2n^2-2n+1}{4n^2-4n+1}\right) - \text{огр. на}$$

$\forall$ компакте сез $(0;+\infty)$

$\Rightarrow \tilde{\theta}_2'$ - регулярна.

Неравенство Крамера-Рао:

$$D\tilde{\theta} \geq \frac{1}{nI(\theta)}$$

$$D\tilde{\theta}_1 \geq \frac{1}{3\cdot\frac{1}{\theta^2}} = \frac{\theta^2}{3} = D\tilde{\theta}_1 \Rightarrow \tilde{\theta}_1 \text{ - эффективная оценка (по дост. условию)}$$

$\tilde{\theta}_2' \neq \tilde{\theta}_1$,

$D\tilde{\theta}_2' \neq \frac{\theta^2}{3} \Rightarrow \tilde{\theta}_2'$ - не эффективна

(Если эффективная оценка $\exists$, то она $\exists!$)